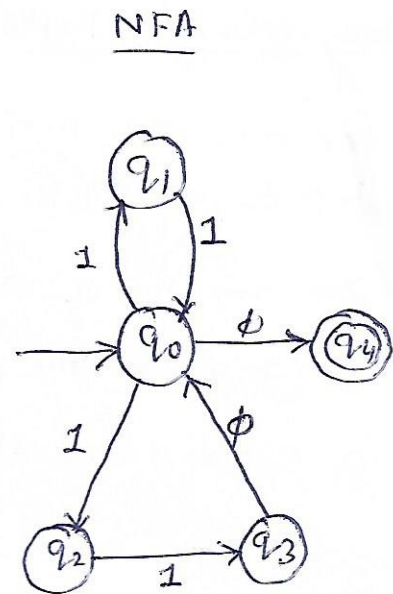
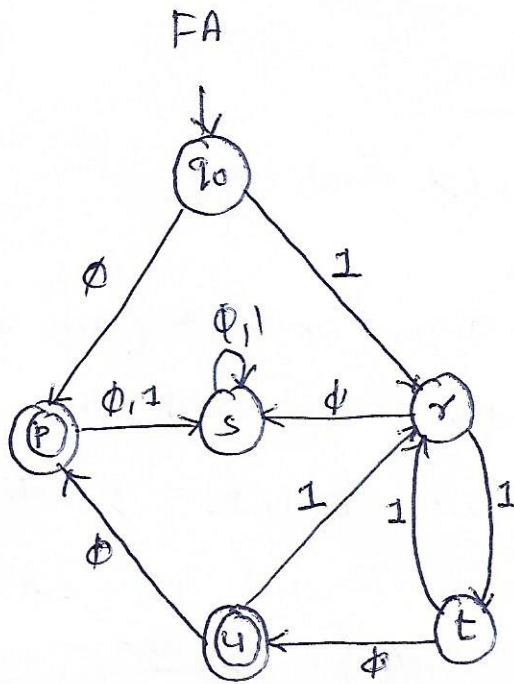


CH 4 — Non-determinism and Kleene's Theorem

①



A simpler approach for accepting $\{11, 110\}^* \{0\}$

NFA

- For some state-input combination, no transition may be specified.
- Multiple transitions from a state for an input symbol. eg. 2 transitions from q_0 on input 1.
- Acceptance of a string : Some path corresponding to the string should lead to an accepting state.
- NFA is allowed to guess and make mistakes.
- Structure of NFA is simple, less states and closely matches the RE.

Definition — [A Non-Deterministic Finite Automaton]

An NFA $M = \langle Q, \Sigma, q_0, A, \delta \rangle$, where Q and Σ are non-empty

finite sets, $q_0 \in Q$, $A \subseteq Q$ and

$$\delta: Q \times \Sigma \rightarrow 2^Q$$

↖ Set of subsets of Q

Similar to FA, δ^* is:

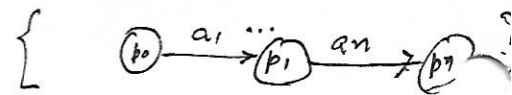
$$\delta^*(p, xa) = \delta(\delta^*(p, x), a)$$

Non-Recursive Defⁿ of δ^* for⁹¹ NFA

For an NFA $M = \langle Q, \Sigma, q_0, A, \delta \rangle$ and any $p \in Q$, $\delta^*(p, \Lambda) = \{p\}$.

For any $p \in Q$ and any $x = a_1 a_2 \dots a_n \in \Sigma^*$ (with $n \geq 1$), $\delta^*(p, x)$ is the set of all states q for which there is a sequence of states $p = p_0, p_1, \dots, p_{n-1}, p_n = q$ satisfying

$$p_i \in \delta(p_{i-1}, a_i) \text{ for each } i \text{ with } 1 \leq i \leq n.$$



Recursive Defⁿ of δ^* for an NFA

Let $M = \langle Q, \Sigma, q_0, A, \delta \rangle$ be an NFA. The function

$\delta^* : Q \times \Sigma^* \rightarrow 2^Q$ is defined as follows:

1. For any $q \in Q$, $\delta^*(q, \Lambda) = \{q\}$
2. For any $q \in Q$, $y \in \Sigma^*$ and $a \in \Sigma$

$$\delta^*(q, ya) = \bigcup_{p \in \delta^*(q, y)} \delta(p, a)$$

Acceptance by an NFA

Let $M = \langle Q, \Sigma, q_0, A, \delta \rangle$ be an NFA.

The string $x \in \Sigma^*$ is accepted by M if

$$\delta^*(q_0, x) \cap A \neq \emptyset$$

Example

State Transition Table for M

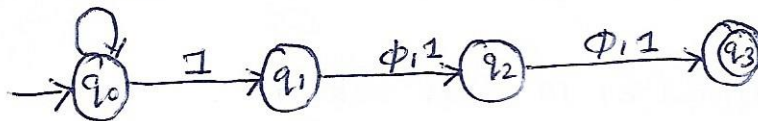
q	$\delta(q, 0)$	$\delta(q, 1)$
q_0	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	$\{q_2\}$
q_2	$\{q_3\}$	$\{q_3\}$
q_3	ϕ	ϕ

$$Q = \{q_0, q_1, q_2, q_3\}$$

$$\Sigma = \{0, 1\}$$

$$A = \{q_3\}$$

State Transition Diagram for M

Let's compute δ^* for some strings using the recursive definition.

i) $\delta^*(q_0, 111)$

ii) $\delta^*(q_0, 011)$

$$\delta^*(q_0, 11) = \bigcup_{p \in \delta^*(q_0, 1)} \delta(p, 1)$$

$$= \bigcup_{p \in \{q_0, q_1\}} \delta(p, 1)$$

$$= \delta(q_0, 1) \cup \delta(q_1, 1)$$

$$= \{q_0, q_1\} \cup \{q_2\}$$

$$= \{q_0, q_1, q_2\}$$

$$\delta^*(q_0, 01)$$

$$= \bigcup_{p \in \delta^*(q_0, 0)} \delta(p, 1)$$

$$= \bigcup_{p \in \{q_0\}} \delta(p, 1)$$

$$= \delta(q_0, 1)$$

$$= \{q_0, q_1\}$$

$$\delta^*(q_0, 111) = \bigcup_{p \in \delta^*(q_0, 11)} \delta(p, 1)$$

$$= \bigcup_{p \in \{q_0, q_1, q_2\}} \delta(p, 1)$$

$$= \delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1)$$

$$= \{q_0, q_1\} \cup \{q_2\} \cup \{q_3\}$$

$$= \{q_0, q_1, q_2, q_3\}$$

$$\begin{aligned}
\delta^*(q_0, 011) &= \bigcup_{p \in \delta^*(q_0, 01)} \delta(p, 1) \\
&= \bigcup_{p \in \{q_0, q_1\}} \delta(p, 1) \\
&= \delta(q_0, 1) \cup \delta(q_1, 1) \\
&= \{q_0, q_1\} \cup \{q_2\} \\
&= \{q_0, q_1, q_2\}
\end{aligned}$$

$\Rightarrow$ 111 is accepted by M but not 011.

The language recognized by M is

$$\begin{aligned}
&\{0, 1\}^* \{1\} \{0, 1\}^* \{0, 1\} \\
&\text{i.e. } \{0, 1\}^* \{1\} \{0, 1\}^* 2 \text{ which is } L_3
\end{aligned}$$

i.e. Set of strings with length atleast 3 and having 1 in the third position from the end.

For any $n \geq 1$, an NFA with $n+1$ states recognizes L_n , whereas an ordinary FA accepting L_n needs atleast 2^n states.

Theorem 4.1

For any NFA $M = \langle Q, \Sigma, q_0, A, \delta \rangle$ accepting language $L \subseteq \Sigma^*$, there is an FA $M_1 = \langle Q_1, \Sigma, q_1, A_1, \delta_1 \rangle$ that also accepts L .

Proof

M_1 is defined as follows:

$$Q_1 = 2^Q \quad q_1 = \{q_0\}$$

$$\text{For } q \in Q_1 \text{ and } a \in \Sigma, \quad \delta_1(q, a) = \bigcup_{r \in q} \delta(r, a)$$

$$A_1 = \{ q \in Q_1 \mid q \cap A \neq \emptyset \}$$

The last defn is the right one because $\{ \text{If a subset has an accepting state, then the subset becomes accepting state in } M_1 \}$ a string x should be accepted by M_1 if, starting in q_0 , the set of states in which M might end up as a result of processing x contains at least one element of A .

The fact that M_1 accepts the same language as M follows from the fact that for any $x \in \Sigma^*$,

$$\delta_1^*(q_1, x) = \delta^*(q_0, x)$$

which we now prove using structural induction on x .

If $x = \lambda$

$$\delta_1^*(q_1, x) = \delta_1^*(q_1, \lambda)$$

$$= q_1 \text{ (by defn of } \delta_1^*)$$

$$= \{q_0\} \text{ (by defn of } q_1)$$

$$= \delta^*(q_0, \lambda) \text{ (by defn of } \delta^*)$$

$$= \delta^*(q_0, x)$$

The induction hypothesis is that x is a string satisfying $\delta_1^*(q_1, x) = \delta^*(q_0, x)$, and we wish to prove that for any $a \in \Sigma$, $\delta_1^*(q_1, xa) = \delta^*(q_0, xa)$:

$$\delta_1^*(q_1, xa) = \delta_1(\delta_1^*(q_1, x), a) \quad (\text{by defn of } \delta_1^*)$$

$$= \delta_1\left(\frac{\delta^*(q_0, x)}{q}, a\right) \quad (\text{by Ind. Hypo.})$$

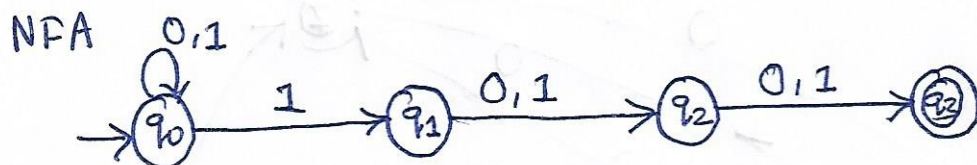
$$\delta_1(q, a) = \bigcup_{x \in q} \delta(x, a) \quad (\text{by our defn of } \delta_1)$$

$$= \delta^*(q_0, xa) \quad (\text{by defn of } \delta^*)$$

That M & M_1 recognize the same lang. is now easy to see. A string x is accepted by M_1 if $\delta_1^*(q_1, x) \in A_1$; we can now say that this is true if and only if $\frac{\delta^*(q_0, x)}{q} \in A_1$; and using the definition of A_1 , we conclude that this is true if and only if $\frac{\delta^*(q_0, x)}{q} \cap A \neq \emptyset$. In other words, x is accepted by M_1 if and only if x is accepted by M .

Applying Subset Construction Algorithm to convert an NFA to an FA

(2^N Subsets Possible for N states of NFA)



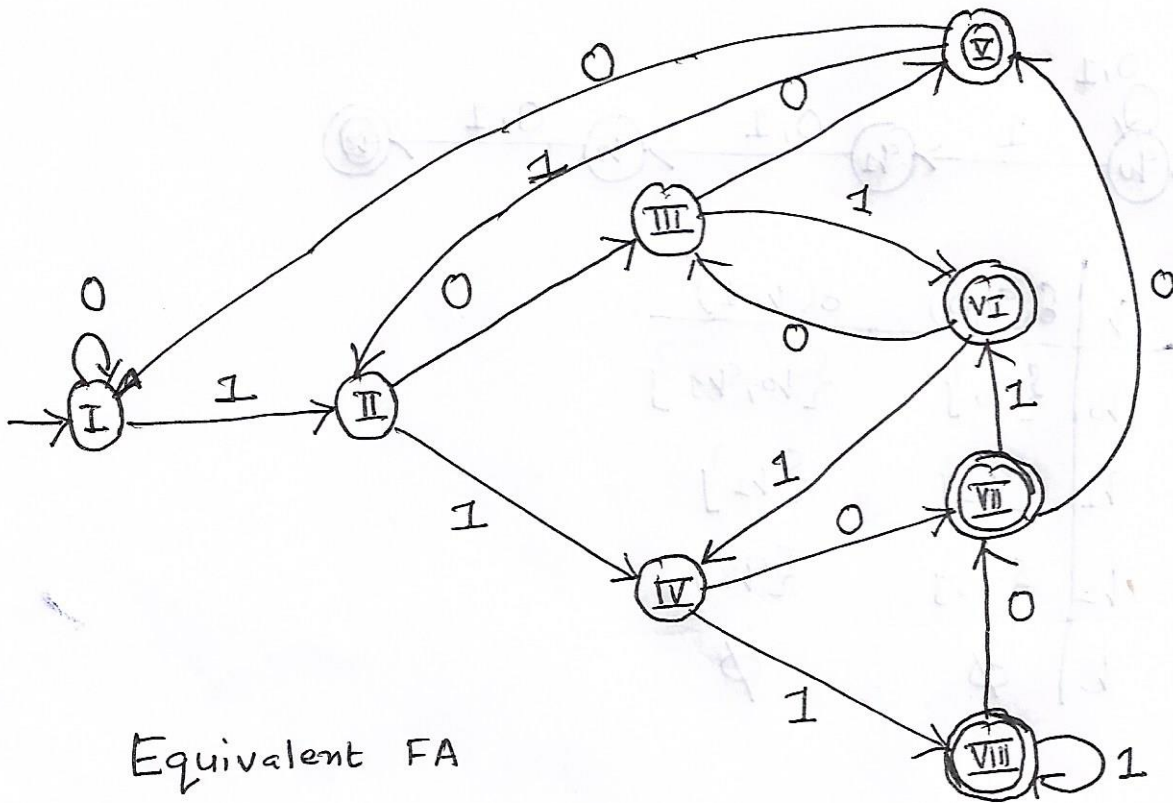
q	$\delta(q, 0)$	$\delta(q, 1)$
q_0	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	$\{q_2\}$
q_2	$\{q_3\}$	$\{q_3\}$
q_3	$\emptyset$	$\emptyset$

FA

Subset s	$\delta_1(s, 0)$	$\delta_1(s, 1)$
$\rightarrow$ I $\{q_0\}$	$\{q_0\}$ I	$\{q_0, q_1\}$ II
II $\{q_0, q_1\}$	$\{q_0, q_2\}$ III	$\{q_0, q_1, q_2\}$ IV
III $\{q_0, q_2\}$	$\{q_0, q_3\}$ V	$\{q_0, q_1, q_3\}$ VI
IV $\{q_0, q_1, q_2\}$	$\{q_0, q_2, q_3\}$ VII	$\{q_0, q_1, q_2, q_3\}$ VIII
V $\{q_0, q_3\}$	$\{q_0\}$ I	$\{q_0, q_1\}$ II
VI $\{q_0, q_1, q_3\}$	$\{q_0, q_2\}$ III	$\{q_0, q_1, q_2\}$ IV
VII $\{q_0, q_2, q_3\}$	$\{q_0, q_3\}$ V	$\{q_0, q_1, q_3\}$ VI
VIII $\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_2, q_3\}$ VII	$\{q_0, q_1, q_2, q_3\}$ VIII

Out of 16 states corresponding to 16 subsets, only 8 are reachable from the initial state.

Therefore, 8 states in the final FA.



Equivalent FA

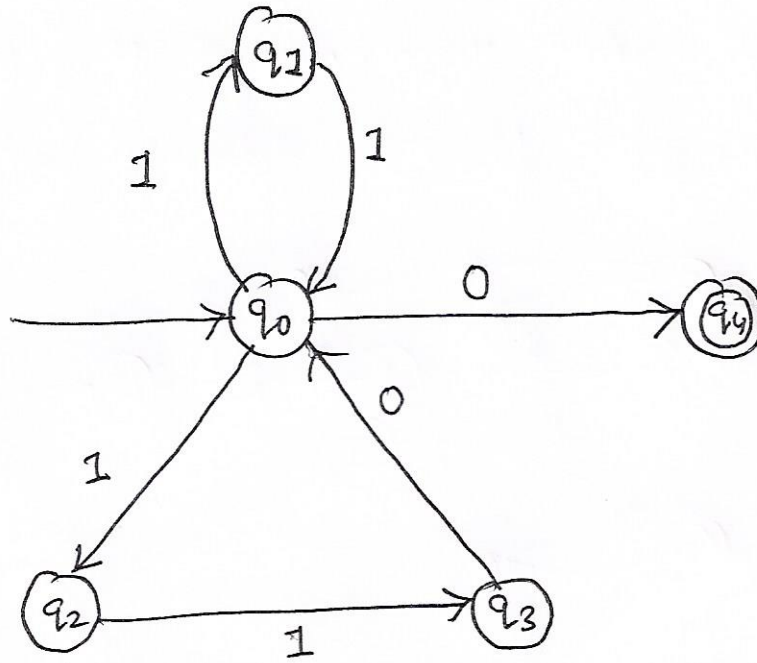
EX. 2 NFA to FA Conversion, using subset construction algorithm

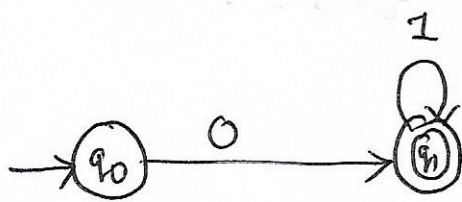
q		$\delta_1(q, 0)$	$\delta_1(q, 1)$
I	$\{q_0\}$	$\{q_4\}$ II	$\{q_1, q_2\}$ III
II	$\{q_4\}$	$\emptyset$ IV	$\emptyset$ IV
III	$\{q_1, q_2\}$	$\emptyset$ IV	$\{q_0, q_3\}$ V
IV	$\emptyset$	$\emptyset$ IV	$\emptyset$ IV
V	$\{q_0, q_3\}$	$\{q_0, q_4\}$ VI	$\{q_1, q_2\}$ III
VI	$\{q_0, q_4\}$	$\{q_4\}$ II	$\{q_1, q_2\}$ III

NFA (S.T. Table)		
q	$\delta(q, 0)$	$\delta(q, 1)$
q_0	q_4	$\{q_1, q_2\}$
q_1	$\emptyset$	$\{q_0\}$
q_2	$\emptyset$	$\{q_3\}$
q_3	$\{q_0\}$	$\emptyset$
q_4	$\emptyset$	$\emptyset$

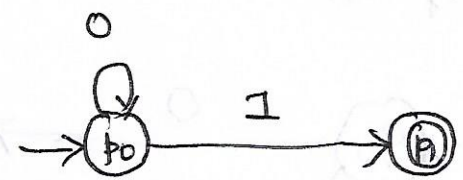
State Transition Diagram of NFA of Ex. 2

8



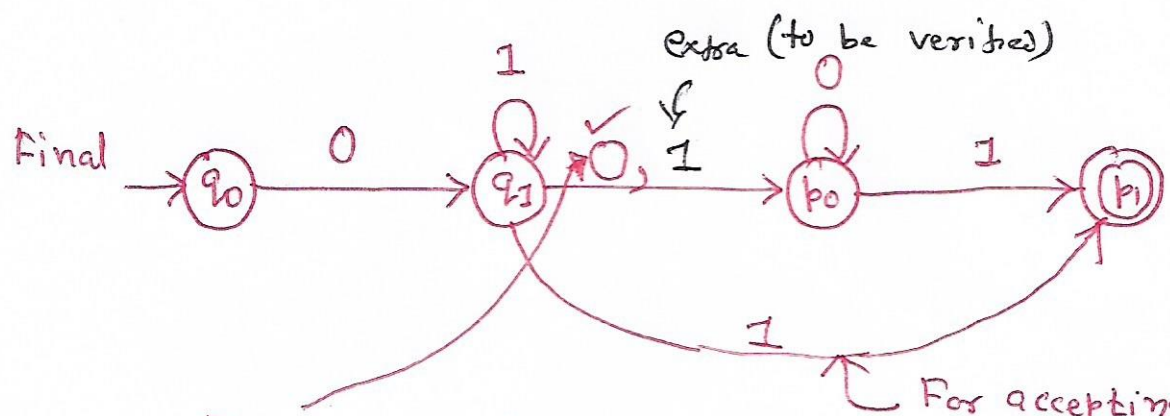
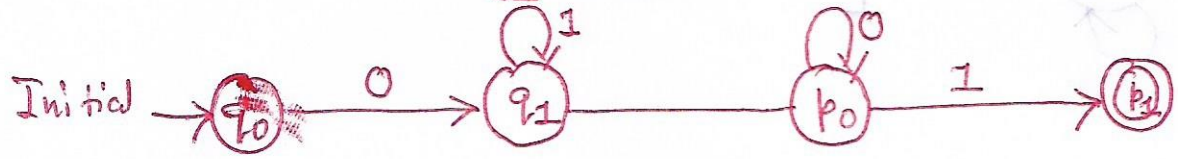


(a) An NFA accepting $\{0\}\{1\}^*$



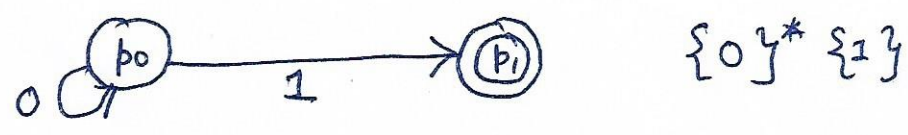
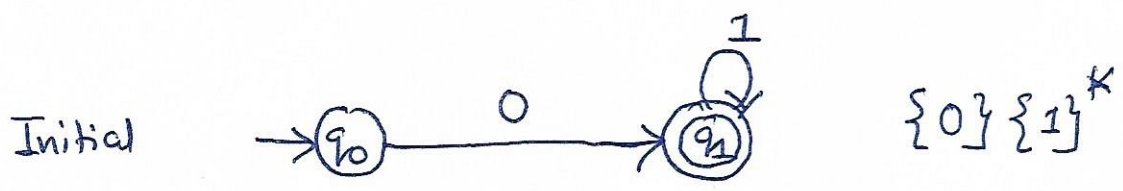
(b) An NFA accepting $\{0\}^*\{1\}$

(c) NFA for $\{0\}\{1\}^*\{0\}^*\{1\}$



Extra (to be verified)
For accepting "001", adding '0' transition in q1
For accepting 01 etc.

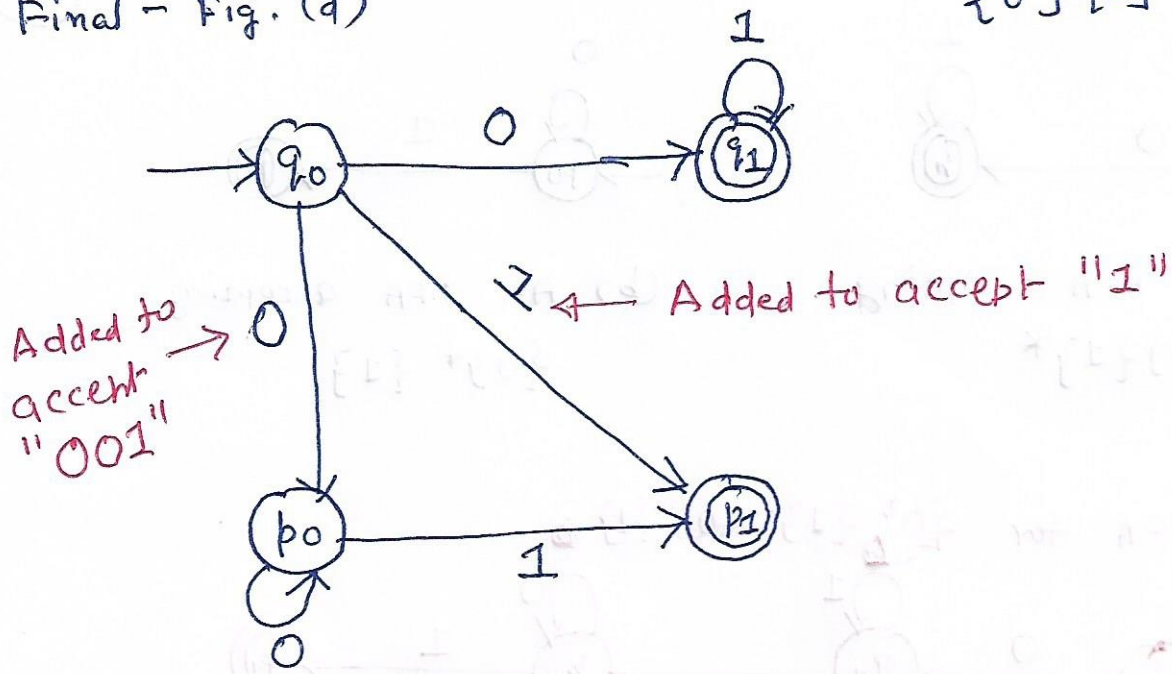
(d) An NFA accepting $\{0\}\{1\}^* \cup \{0\}^*\{1\}$

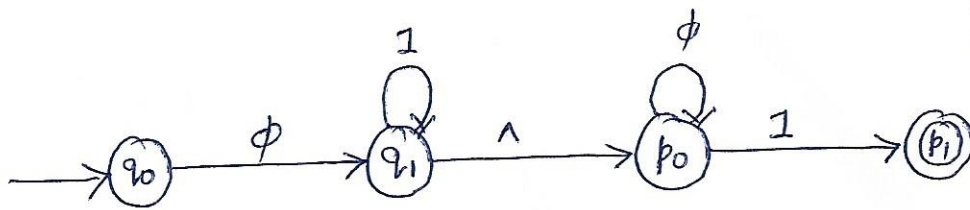


Final - Fig. (d)

$\{0\} \{1\}^* \cup \{0\}^* \{1\}$

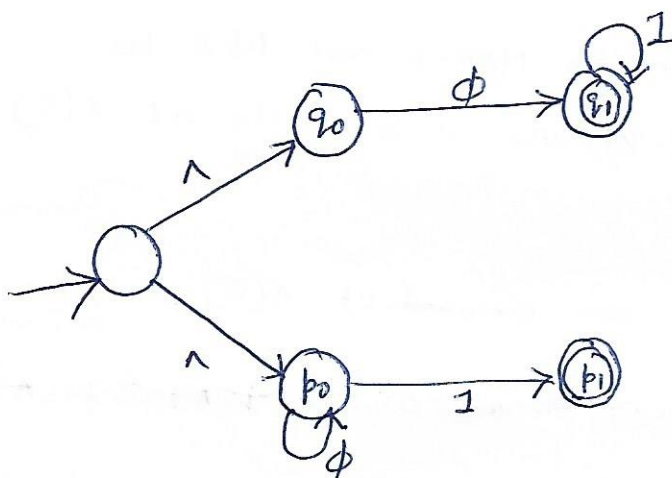
(8c)





→ M guesses in state q_1 that no more i/p for L_1 and proceeds to process L_2 .

(e) An NFA- λ accepting $\{0\}\{1\}^*\{\emptyset\}^*\{1\}$



→ Machine M makes guess and operates like M_1 or M_2

(f) An NFA- λ accepting $\{0\}\{1\}^* \cup \{0\}^*\{1\}$

⇒ NFA- λ are more general than NFAs as they are allowed to make transitions, not only on input symbols from the alphabet, but also on null inputs.

* NFA- λ is a 5-tuple $(Q, \Sigma, q_0, A, \delta)$, where Q and Σ are finite ~~states~~ sets, $q_0 \in Q$, $A \subseteq Q$ and

$$\delta: Q \times (\Sigma \cup \{\lambda\}) \rightarrow 2^Q$$

Non-Recursive defⁿ of δ^* for NFA- λ

Very similar to NFA.

λ -closure of a set of states

Let $M = \langle Q, \Sigma, q_0, A, \delta \rangle$ be an NFA- λ and let S be any subset of Q . The λ -closure of S is the set $\lambda(S)$ defined as follows:

1. Every element of S is an element of $\lambda(S)$.
2. For any $q \in \lambda(S)$, every element of $\delta(q, \lambda)$ is in $\lambda(S)$.
3. No other elements of Q are in $\lambda(S)$.

Recursive defⁿ of δ^* for an NFA- λ

Let $M = \langle Q, \Sigma, q_0, A, \delta \rangle$ be an NFA- λ .

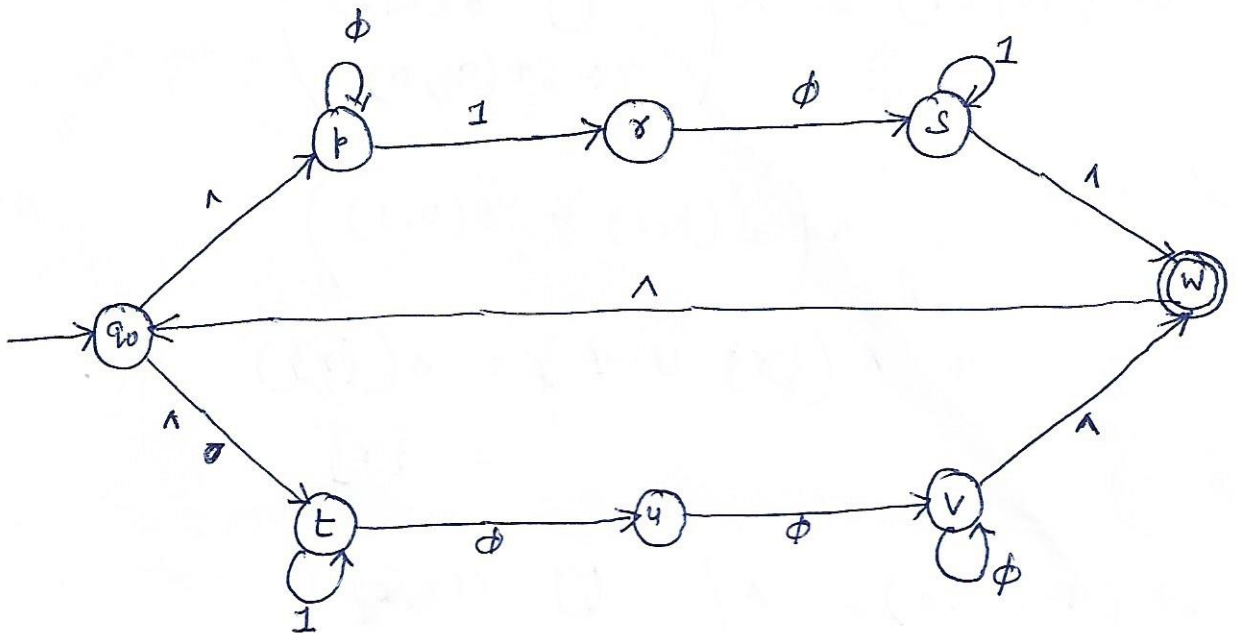
The $\delta^* : Q \times \Sigma^* \rightarrow 2^Q$ is defined as follows:

1. For any $q \in Q$, $\delta^*(q, \lambda) = \lambda(\{q\})$
2. For any $q \in Q$, $y \in \Sigma^*$ and $a \in \Sigma$,

$$\delta^*(q, ya) = \lambda \left(\bigcup_{r \in \delta^*(q, y)} \delta(r, a) \right)$$

$\rightarrow$ A string x is accepted by M if $\delta^*(q_0, x) \cap A \neq \emptyset$

Can we write $\lambda(A)$ here?
No. Because, it could lead to
multiple states as well.



$$\begin{aligned} \delta^*(q_0, 1) &= 1(\{q_0\}) \\ &= \{q_0, p, t\} \end{aligned}$$

$$\delta^*(q_0, 0) = 1 \left(\bigcup_{z \in \delta^*(q_0, 1)} \delta(z, 0) \right)$$

$$= 1 \left(\bigcup_{z \in \{q_0, p, t\}} \delta(z, 0) \right)$$

$$= 1 \left(\delta(q_0, 0) \cup \delta(p, 0) \cup \delta(t, 0) \right)$$

$$= 1 \left(\phi \cup \{p\} \cup \{u\} \right)$$

$$= 1 \left(\{p, u\} \right)$$

$$= \{p, u\}$$

$$\delta^*(q_0, 01) = \bigwedge \left(\bigcup_{z \in \delta^*(q_0, 0)} \delta(z, 1) \right)$$

$$= \bigwedge \left(\delta(p, 1) \cup \delta(u, 1) \right)$$

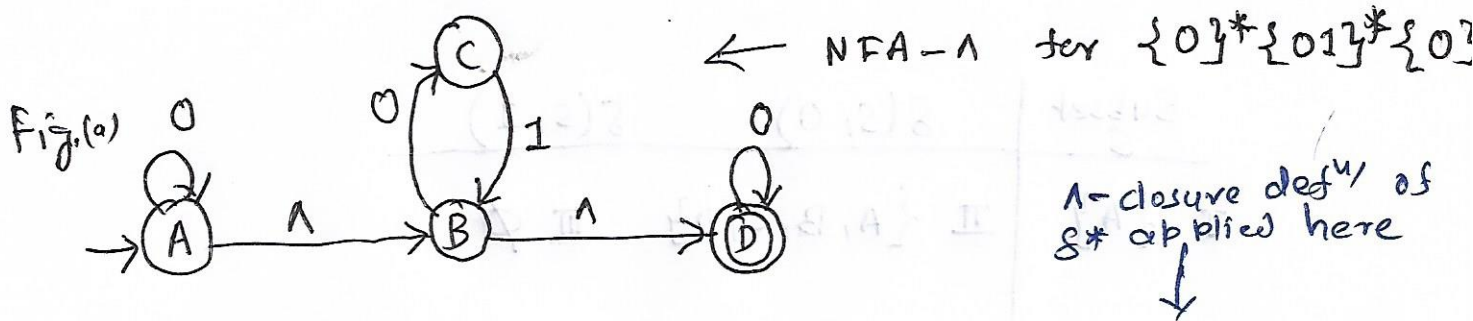
$$= \bigwedge (\{x\} \cup \emptyset) = \bigwedge (\{x\}) \\ = \{x\}$$

$$\delta^*(q_0, 010) = \bigwedge \left(\bigcup_{z \in \delta^*(q_0, 01)} \delta(z, 0) \right)$$

$$= \bigwedge \left(\delta(x, 0) \right)$$

$$= \bigwedge (\{s\}) = \cancel{\{s\}} \{s, w, q_0, p, t\}$$

Converting NFA- Λ to an NFA



q	$\delta(q, \Lambda)$	$\delta(q, 0)$	$\delta(q, 1)$	$\delta^+(q, 0)$	$\delta^+(q, 1)$
A	$\{B\}$	$\{A, B\}$	$\emptyset$	$\{A, B, C, D\}$	$\emptyset$
B	$\{D\}$	$\{C\}$	$\emptyset$	$\{C, D\}$	$\emptyset$
C	$\emptyset$	$\emptyset$	$\{B\}$	$\emptyset$	$\{B, D\}$
D	$\emptyset$	$\{D\}$	$\emptyset$	$\{D\}$	$\emptyset$

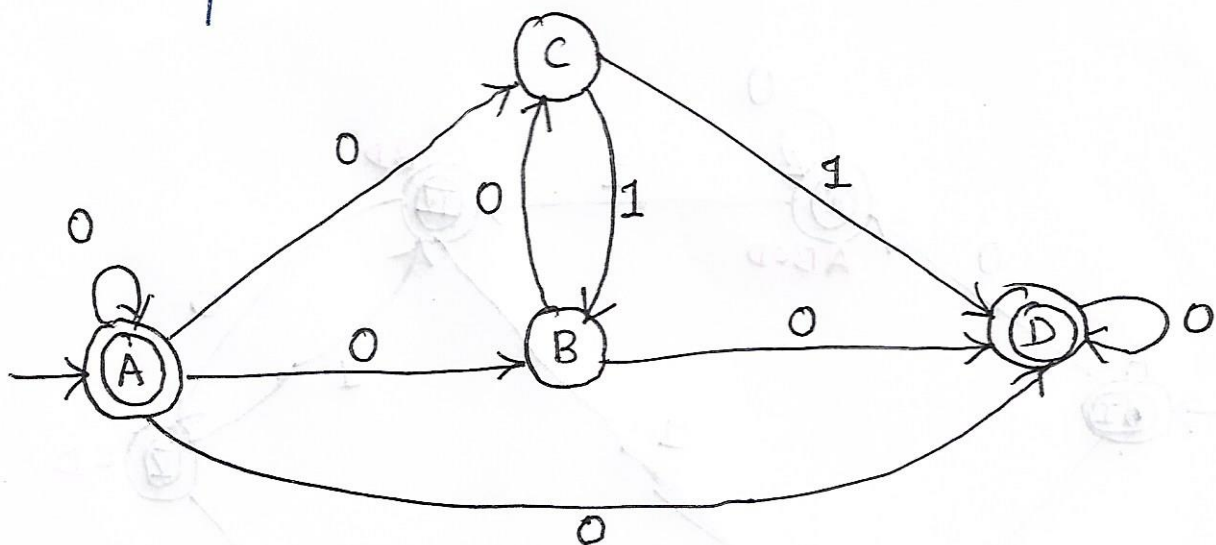
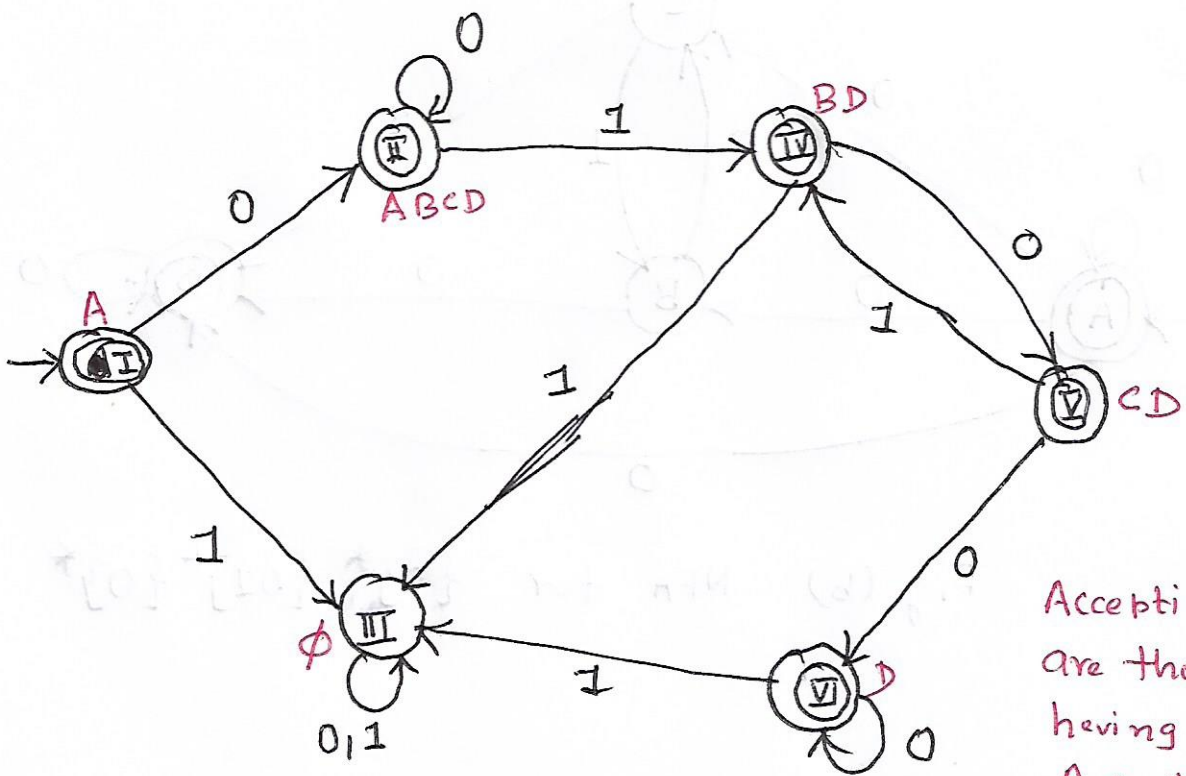


Fig. (b) NFA for $\{0\}^* \{01\}^* \{0\}^*$

Subset S	$\delta(S, 0)$	$\delta(S, 1)$
I $\{A\}$	II $\{A, B, C, D\}$	III $\emptyset$
II $\{A, B, C, D\}$	II $\{A, B, C, D\}$	IV $\{B, D\}$
III $\emptyset$	III $\emptyset$	III $\emptyset$
IV $\{B, D\}$	V $\{C, D\}$	III $\emptyset$
V $\{C, D\}$	VI $\{D\}$	IV $\{B, D\}$
VI $\{D\}$	VI $\{D\}$	III $\emptyset$



Accepting states
are those
having either
A or D in the
subset

An NFA for $\{0\}^* \{01\}^* \{0\}^*$

Another example for converting NFA-1 to NFA

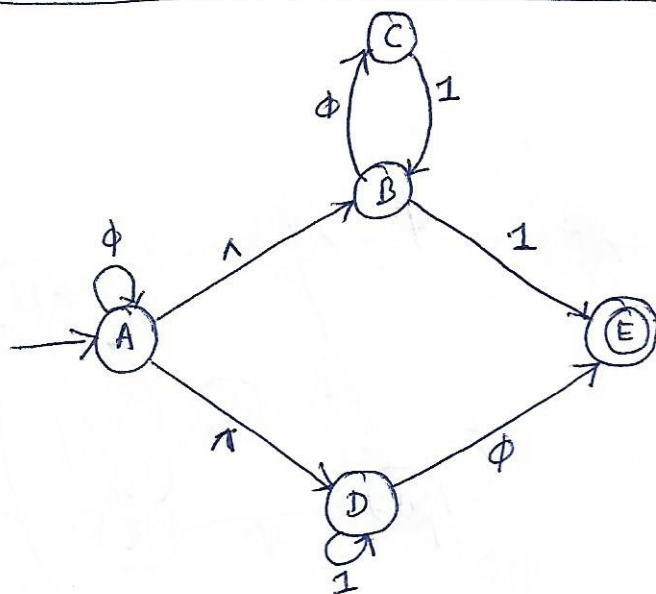


Fig. (a)

NFA-1 for

$$\{\phi\}^* \{\phi 1\}^* \{1\}$$

$$\cup \{1\}^* \{0\}$$

~~010~~

010 is accepted
but not belong to
the language if
we omit.

q	$\delta(q, 1)$	$\delta(q, 0)$	$\delta(q, 1)$	$\delta^*(q, 0)$	$\delta^*(q, 1)$
A	{B, D}	{A}	ϕ	{A, B, C, D, E}	{D, E}
B	ϕ	{C}	{E}	{C}	{E}
C	ϕ	ϕ	{B}	ϕ	{B}
D	ϕ	{E}	{D}	{E}	{D}
E	ϕ	ϕ	ϕ	ϕ	ϕ

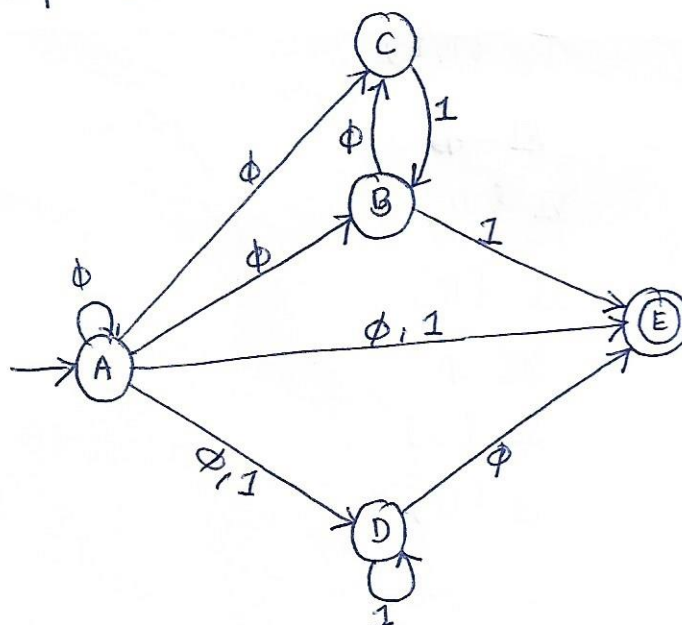
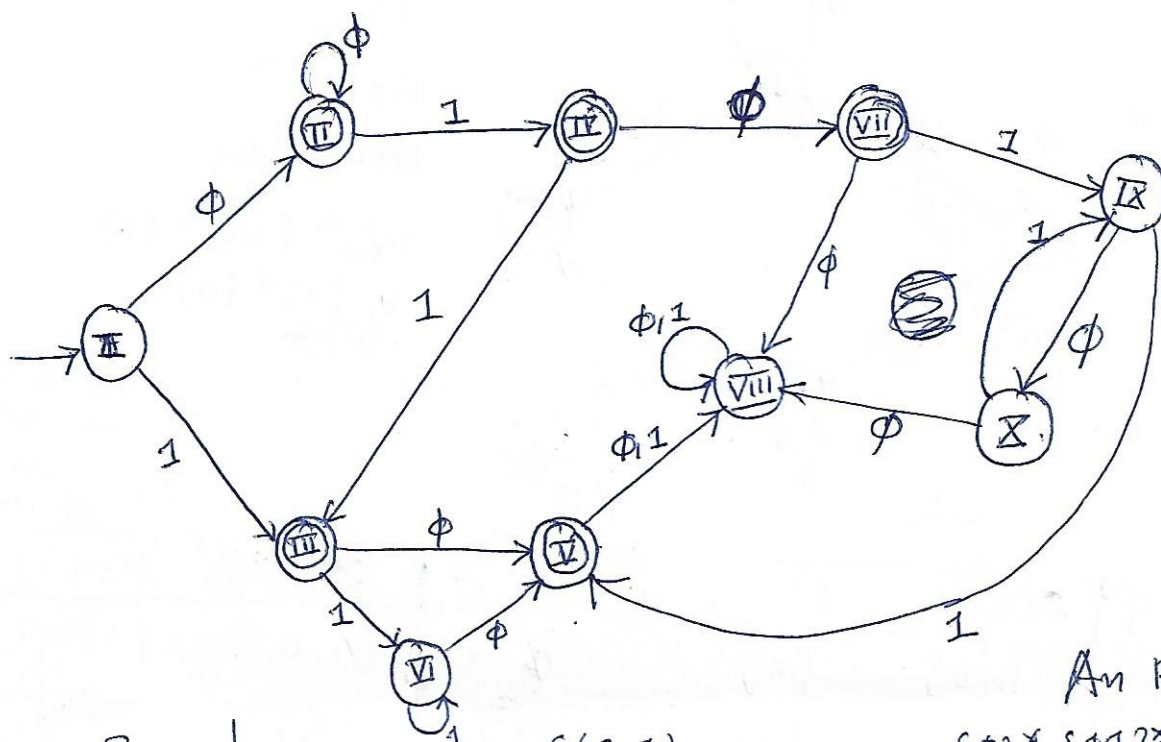


Fig. (b)

NFA For above Lang.



An FA for
 $\{0\}^* \{01\}^* \{1\} \cup$
 $\{0\}^* \{1\}^* \{0\}$

Subsets	$\delta(S, 0)$	$\delta(S, 1)$
I {A}	II {A, B, C, D, E}	III {D, E}
II {A, B, C, D, E}	II {A, B, C, D, E}	IV {B, D, E}
III {D, E}	V {E}	VI {D}
IV {B, D, E}	VII {C, E}	III {D, E}
V {E}	VIII $\emptyset$	VIII $\emptyset$
VI {D}	V {E}	VI {D}
VII {C, E}	VII $\emptyset$	IX {B}
VIII $\emptyset$	VIII $\emptyset$	VIII $\emptyset$
IX {B}	X {C}	V {E}
X {C}	VII $\emptyset$	IX {B}

Kleene's Theorem - Part I

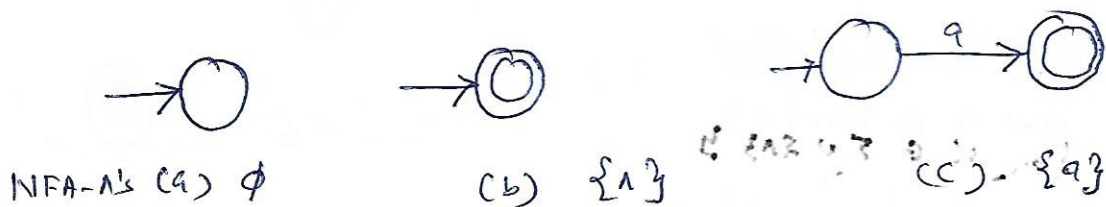
Any Regular Language can be accepted by a finite automaton.

We will prove this using the structural induction.

(~~The~~ Reg. Lang. are closed under union, concatenation and Kleene's $*$)

The basic step of the proof is to show that the three basic languages can be accepted by NFA-ns.

i.e. ϕ , $\{1\}$, $\{a\}$ for $a \in \Sigma$.



Induction Hypo. : L_1 and L_2 are languages that can be accepted by NFA-ns.

Induction step: $L_1 \cup L_2$, $L_1 L_2$, L_1^* can be accepted by some NFA-ns.

- Suppose that L_1 and L_2 are recognized by NFA-ns M_1 and M_2 respectively, i.e. $M_i = (Q_i, \Sigma, q_i, A_i, \delta_i)$ for $i=1, 2$.
- By renaming states, if necessary, we assume that $Q_1 \cap Q_2 = \phi$.
- We will construct NFA-ns M_u , M_c , M_k for recognizing the languages $L_1 \cup L_2$, $L_1 L_2$, L_1^* respectively.

Construction of $M_4 = (Q_4, \Sigma, q_4, A_4, \delta_4)$,

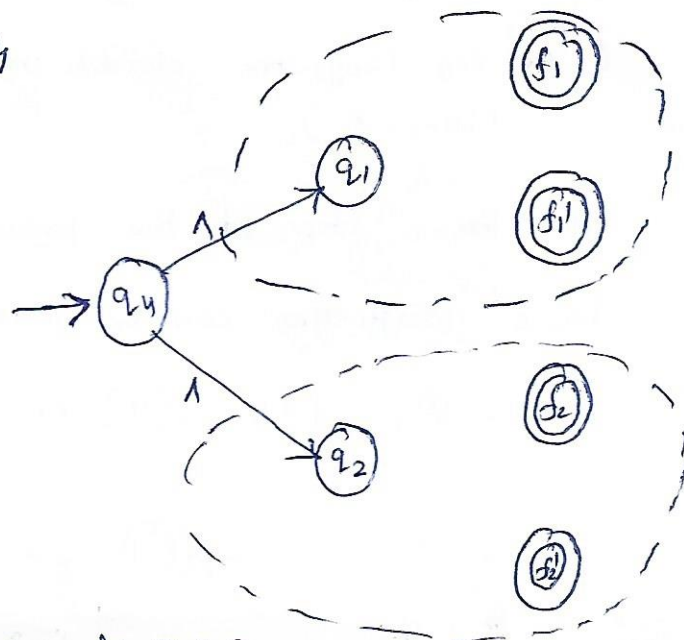
Let q_4 be a new state not in either Q_1 or Q_2 .

$$Q_4 = Q_1 \cup Q_2 \cup \{q_4\}$$

$$A_4 = A_1 \cup A_2$$

$$\delta_4(q_4, \lambda) = \{q_1, q_2\}$$

$$\delta_4(q_4, a) = \phi \text{ for every } a \in \Sigma$$



And for each $q \in Q_1 \cup Q_2$

$$\delta_4(q, a) = \begin{cases} \delta_1(q, a) & \text{if } q \in Q_1 \\ \delta_2(q, a) & \text{if } q \in Q_2 \end{cases}$$

and $a \in \Sigma \cup \{\lambda\}$
 {or} $a \in \Sigma \cup \{\lambda\}$

To prove that M_4 accepts $L_1 \cup L_2$

i) if $x \in L_1 \cup L_2$, M_4 accepts it.

M_4 moves to q_1 or q_2 from q_4 through λ -transition.

Then process x the way M_1 or M_2 would do.

Thus x would be accepted by M_4 .

ii) If M_4 accepts x , then $x \in L_1 \cup L_2$

As M_4 accepts x , there is a sequence of transition ^{for x} from q_4 to one of the accepting states of either M_1 or M_2 .

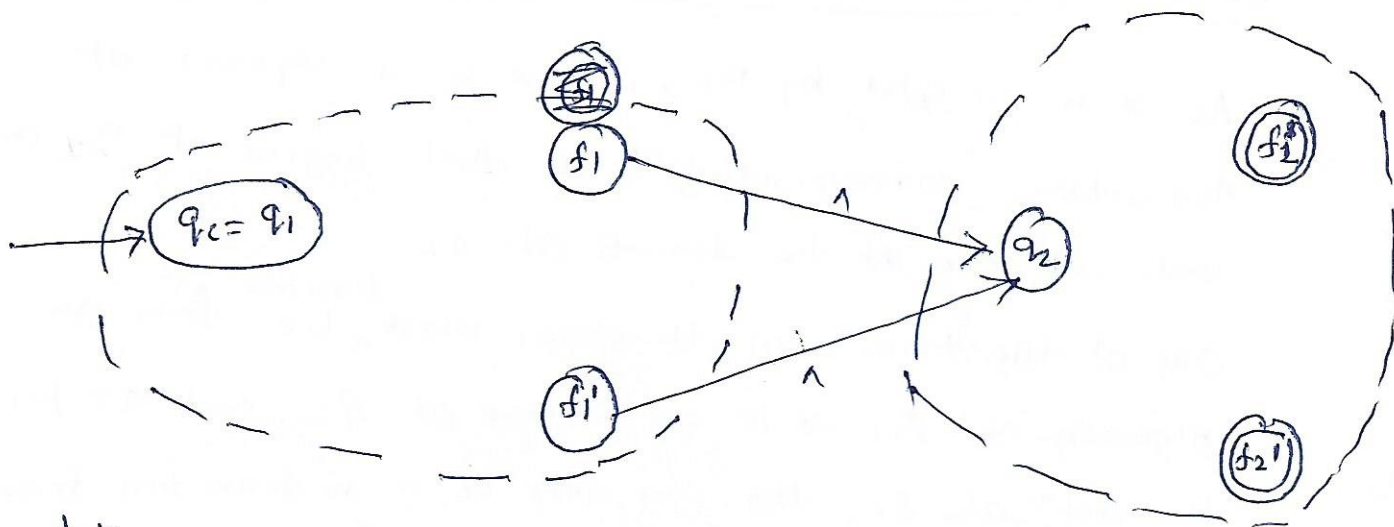
The first of transition must be to q_1 or q_2 from q_4 on λ as no other transition is present from q_4 .

Thereafter, all other transitions are from q_1 to elements of Q_1 or from q_2 to elements of Q_2 , as $Q_1 \cap Q_2 = \phi$

Therefore x is accepted either by M_1 or M_2 .

$\therefore x \in L_1 \cup L_2$

Construction of $M_c = (Q_c, \Sigma, q_c, A_c, \delta_c)$



Let

$$Q_c = Q_1 \cup Q_2 \cup \{q\}$$

$$q_c = q_1$$

$$A_c = A_2$$

δ_c

$$\rightarrow \text{or } \{q \in Q_1 \cup Q_2 - A_1\}$$

For any q not in A_1 and $\alpha \in \Sigma \cup \{1\}$

Case I $q \in Q_1 \cup Q_2 - A_1$

$$\text{i.e., } \delta_c(q, \alpha) = \begin{cases} \delta_1(q, \alpha) & \text{if } q \in Q_1 - A_1 \\ \delta_2(q, \alpha) & \text{if } q \in Q_2 \end{cases}$$

Case II $q \in A_1$

For $q \in A_1$

$$\delta_c(q, a) = \delta_1(q, a) \text{ for every } a \in \Sigma$$

$$\delta_c(q, 1) = \delta_1(q, 1) \cup \{q_2\}$$

Proof

If $x \in L_1 L_2$, M_c would accept it

On i/p string $x_1 x_2$, where $x_i \in L_i$ for both values of i , M_c can process x_1 , arriving at a state in A_1 ; jump from this state to q_2 by a 1 -transition; and

then process x_2 the way M_2 would, so that $x_1 x_2$ is accepted.

Conversely, if x is accepted by M_C then $x \in L_1 L_2$.

As x is accepted by M_C , there is a sequence of transitions corresponding to x that begins at q_1 and ends at one of the elements of A_2 .

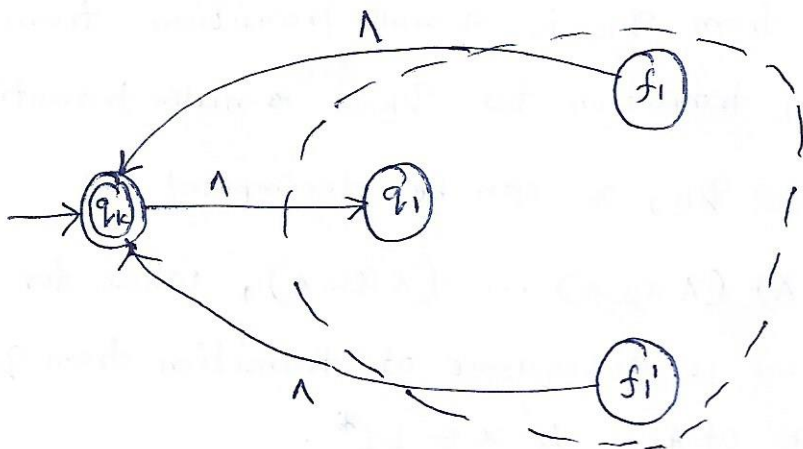
One of the transition, ~~therefore~~, must ^{therefore} be from an element of Q_1 to an element of Q_2 , and as per the defⁿ of δ_C , this can only be a λ -transition from an element of A_1 to q_2 .

Because $Q_1 \cap Q_2 = \emptyset$, all the previous transitions are from q_1 to element of A_1 and all the subsequent transitions are from q_2 to ^{any} elements of A_2 .

It follows that $x = x_1 \wedge x_2 = x_1 x_2$, where x_1 is accepted by M_1 and x_2 is accepted by M_2 .

$\therefore x \in L_1 L_2$

Construction of M_k :



$$M_k = (Q_k, \Sigma, q_k, A_k, \delta_k)$$

$$Q_k = Q_1 \cup \{q_k\}$$

$$A_k = \{q_k\}$$

$$\delta_k(q_k, \lambda) = \{q_1\}$$

$$\delta_k(q_k, a) = \emptyset \text{ for } a \in \Sigma$$

{Simple defn of δ_k }

(A) For $q \in Q_1 - A_1$
and $a \in \Sigma \cup \{\lambda\}$

$$\delta_k(q, a) = \delta_1(q, a)$$

(B) For $q \in A_1$

$$\delta_k(q, a) = \delta_1(q, a) \text{ for } a \in \Sigma$$

$$\delta_k(q, \lambda) = \delta_1(q, \lambda) \cup \{q_k\}$$

Use

(A)

instead
of this

For $q \in Q_1$ and $a \in \Sigma \cup \{\lambda\}$,

$$\delta_k(q, a) = \delta_1(q, a) \text{ except when } q \in A_1 \text{ and } a \text{ is } \lambda.$$

(B)

For $q \in A_1$ and a is λ

$$\delta_k(q, \lambda) = \delta_1(q, \lambda) \cup \{q_k\}$$

To Prove: If $x \in L_1^*$, then M_k accepts it.

Proof: If x is λ , Then it is accepted by M_k .

Otherwise for $m \geq 1$, $x = x_1 x_2 \dots x_m$ where each $x_i \in L_1$

M_k then makes a null-transition from q_k to q_1 and then process x_i like M_1 and take a λ -transition from an element of A_1 to q_k .

$\therefore x = (\lambda x_1 \lambda) (\lambda x_2 \lambda) \dots (\lambda x_m \lambda)$ is accepted by M_k , (where each $x_i \in L_1$)

Converse, if $x \in L_1$ is accepted by M_k , then $x \in L_1^*$.

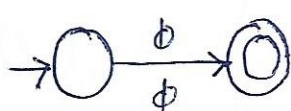
Because x is accepted by M_k , there is a sequence of transition corresponding to x that begins and ends at q_k .

Since, only transition from q_k is a null-transition from q_k to q_1 and only transition to q_k ^{are} ~~is~~ null-transitions from elements of A_1 to q_k , x can be decomposed as:

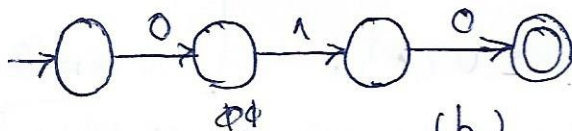
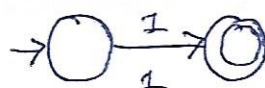
$x = (\lambda x_1 \lambda) (\lambda x_2 \lambda) \dots (\lambda x_m \lambda)$, where for each x_i , there is a sequence of transition from q_1 to an element of A_1 . $\therefore x \in L_1^*$.

Constructing an NFA- λ for $(00+1)^* (10)^*$.

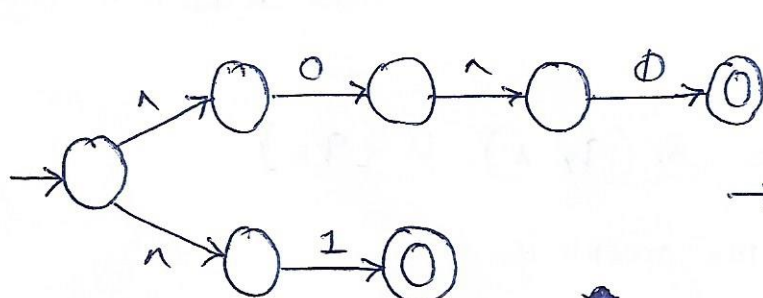
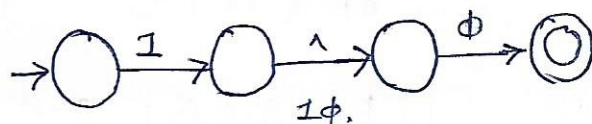
(Using the Algo.)



(a)

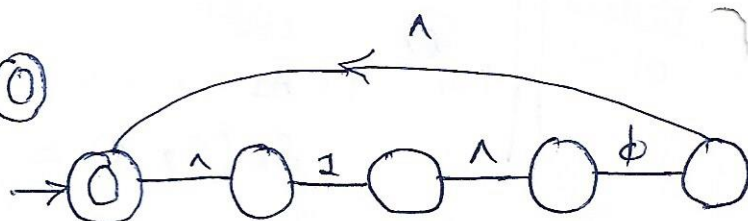


(b)

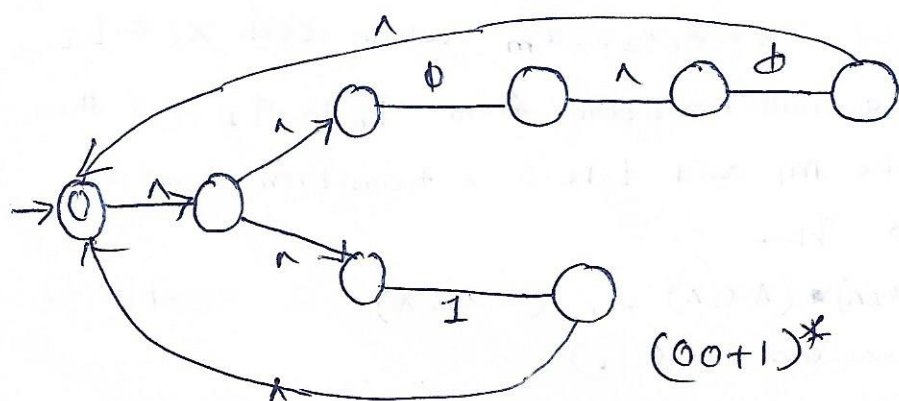


(c)

$(\phi\phi+1)^*$



$(1\phi)^*$



$(00+1)^*$

Converse, if $x \in L_1$ is accepted by M_k , then $x \in L_1^*$.

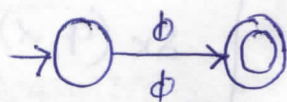
Because x is accepted by M_k , there is a sequence of transition corresponding to x that begins and ends at q_k .

Since, only transition from q_k is a null-transition from q_k to q_1 and only transition to q_k ^{are} ~~is~~ null-transitions from elements of A_1 to q_k , x can be decomposed as:

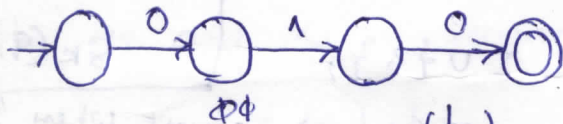
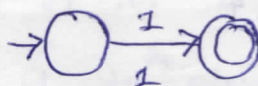
$x = (\lambda x_1 \lambda) (\lambda x_2 \lambda) \dots (\lambda x_m \lambda)$, where for each x_i , there is a sequence of transition from q_1 to an element of A_1 . $\therefore x \in L_1^*$.

Constructing an NFA- λ for $(00+1)^* (10)^*$.

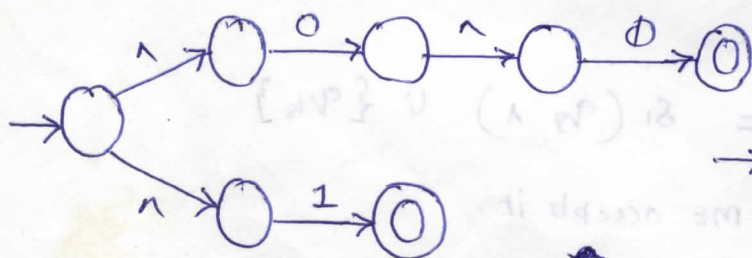
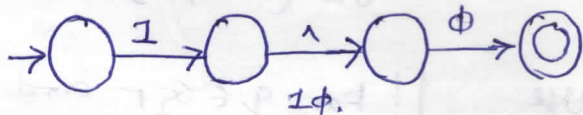
(Using the Algo.)



(a)

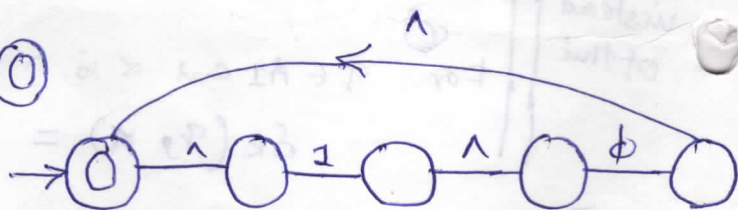


(b)

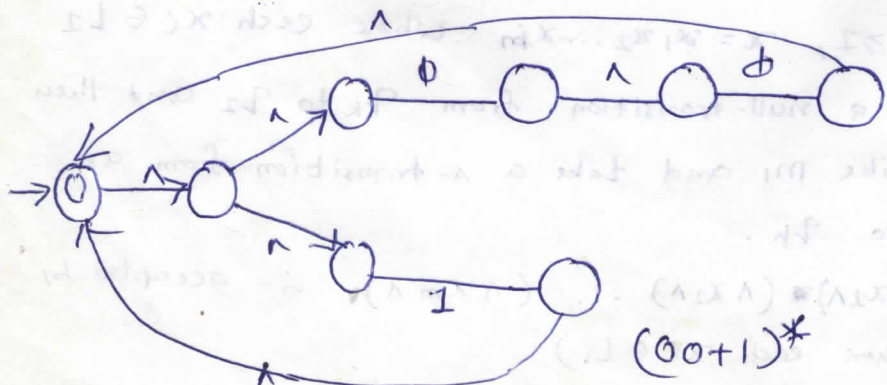


(c)

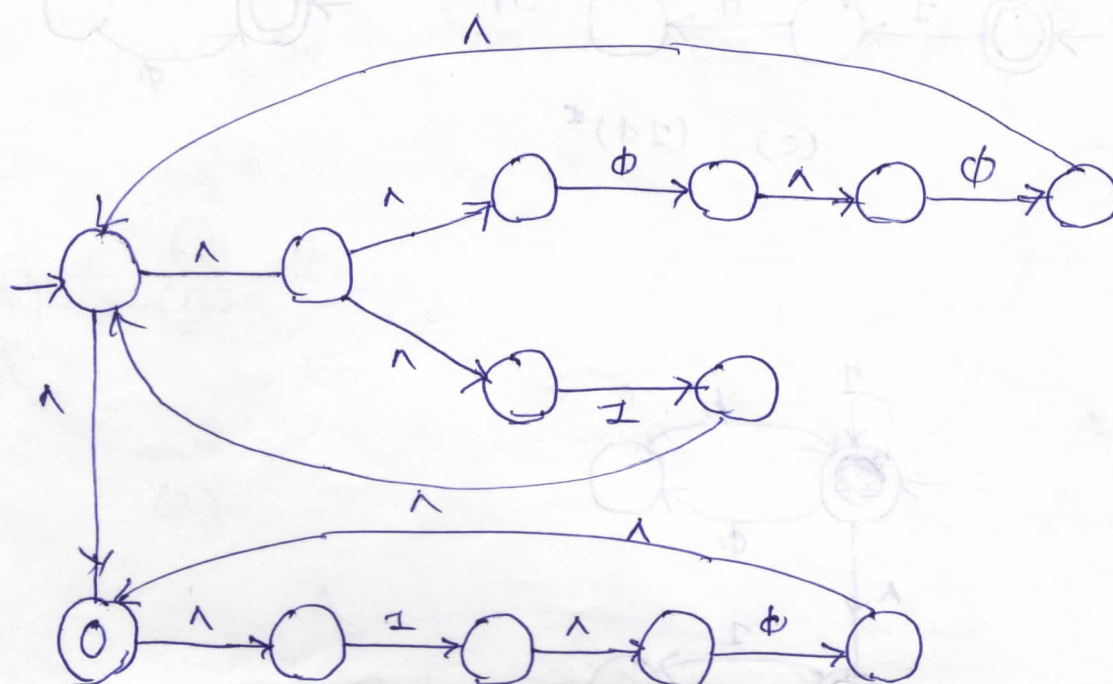
$(00+1)^*$



$(10)^*$

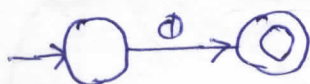


$(00+1)^* (10)^*$



$(00+1)^* (10)^*$

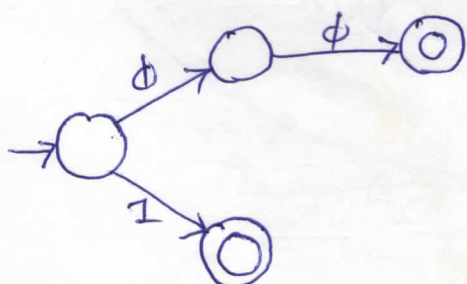
Simplified Algo NFA-1 for $(00+1)^* (10)^*$



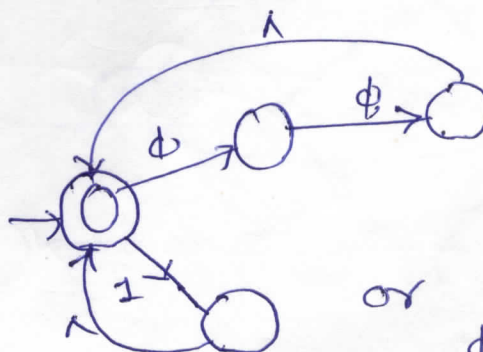
(a)



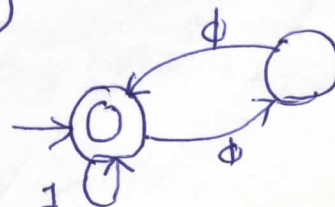
(b)



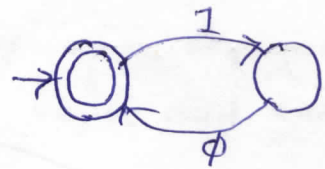
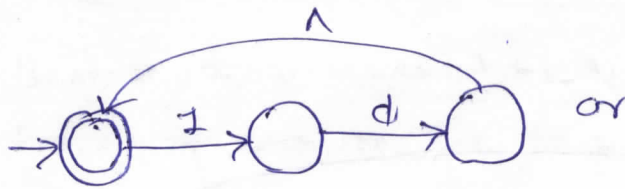
(c) $(00+1)$



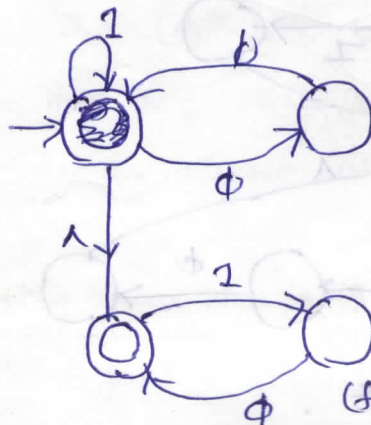
or



(e) $(00+1)^*$



(e) $(1\phi)^*$



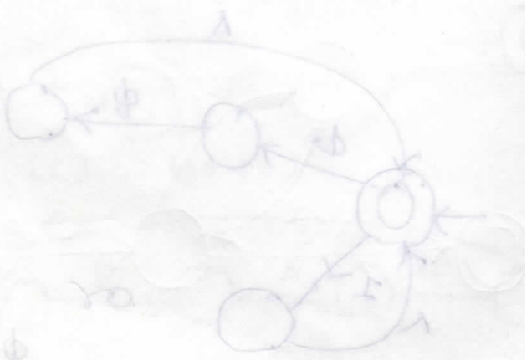
(f) $(00+1)^* (1\phi)^*$



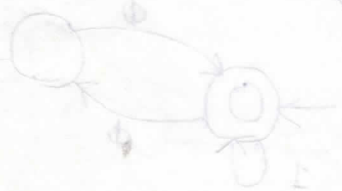
(c)



(d)



(1+φφ) (1)



$(1+φφ)^* (1b)$

Ex. 3

Draw NFA-1 for $(0+1)(01)^*(011)^*$.
Simplified version is acceptable.

Fig. (a) $(0+1)(01)^*$

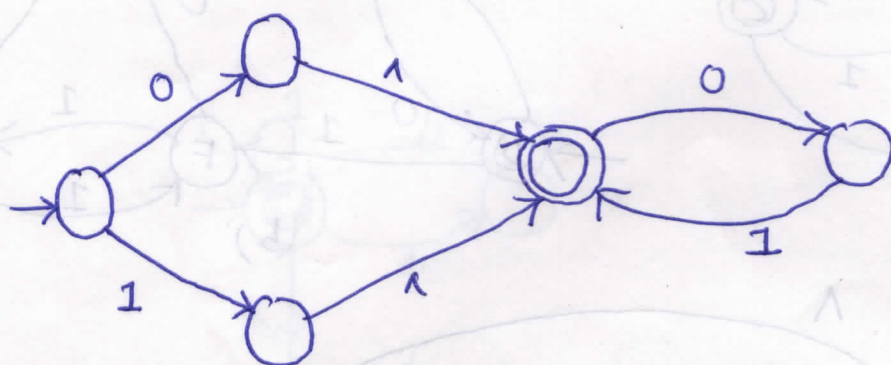


Fig. (b) $(0+1)(01)^*(011)^*$

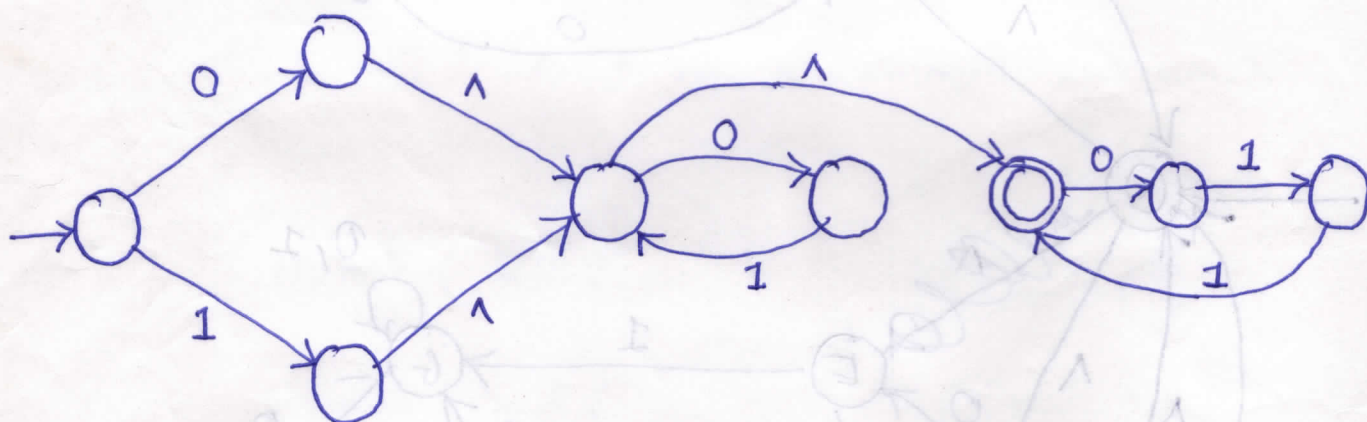


Fig. (c) $(0+1)(01)^*(011)^*$ — Simplified

